

Data for Policy Analysis and Management

CLASS 8: EXPERIMENTAL AND
QUASI-EXPERIMENTAL
METHODS INTRODUCTION

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Agenda

1. Correlation vs. Causation
2. Experimental Design
3. Quasi-experimental designs
4. Case study: MTO
5. Choosing the best design
6. Presentations

Introduction to Experimental and Quasi-Experimental Methods

- Understanding causal relationships is at the heart of social science research, whether studying interventions, policies, or exposures.

Why Causal Methods Matter

- Correlations are easy to observe, but causation requires credible comparison.
- Key questions: Do neighborhoods affect outcomes? Do workforce programs raise wages? Do policies change behavior?
- Experimental and quasi-experimental designs help estimate causal effects beyond simple comparisons.



From Correlation to Causation

1

We want to estimate the **counterfactual**: what would have happened to the individual without the treatment or intervention?

2

Treatment effect = outcome with treatment – outcome without treatment. **This difference captures the true causal impact.**

3

The core problem: **only one outcome is ever observed** per individual → treated or untreated, never both.

*Research designs attempt to **approximate** the missing counterfactual (the outcome we can never directly observe).*

Experimental Designs: The Gold Standard

Introduce
Intervention

Random
Assignment

Compare
Outcomes

- Researchers introduce a controlled intervention into a real-world or lab setting to test its effect on participants.

- Participants are randomly assigned to either a treatment group (receives intervention) or a control group (does not).
- Randomization ensures both groups are similar on average across observed and unobserved characteristics.
- This eliminates selection bias and makes outcome differences attributable to the treatment.

- Outcomes are compared between groups.
- Example: families randomly assigned to receive housing vouchers vs. those who did not, measuring neighborhood and wellbeing outcomes.

Strengths and Limitations of Experiments

Why Experiments Work

- Strong internal validity --> results reflect true causal effects.
- Reduces selection bias through random assignment.
- Provides clear causal interpretation of outcomes.
- Allows estimation of average treatment effects across the study population.

Key Challenges

- Ethical concerns and high costs can make randomization difficult or impossible.
- External validity and generalizability to other contexts may be limited.
- Recruitment difficulties and long time horizons add complexity.
- Compliance issues arise when participants do not follow their assigned treatment.

Quasi-Experimental Designs: When Randomization Is Not Possible

- When randomization is not feasible, researchers rely on natural variation (such as differences in policies, timing, or geography) to construct credible comparisons.
- The core goal is to **identify comparison groups that approximate what the treated group would have experienced without the intervention**, thereby estimating the counterfactual.
- These designs exploit real-world circumstances rather than researcher-controlled assignment, making them practical for studying social programs and policy interventions.
- Capili & Anastasi (2024) emphasize that quasi-experimental designs, when carefully applied, can yield causal estimates that closely rival those from randomized experiments.



Common Quasi-Experimental Designs

Difference-in-Differences (DiD)

- Compares pre-post changes in outcomes between a treatment group and a comparison group over time.
- The key assumption is that both groups would have followed parallel trends without the intervention.

Example: One state adopts a new childcare subsidy policy while a neighboring state does not. DiD estimates the policy effect by comparing changes in childcare enrollment across both states before and after adoption.

Regression Discontinuity Design (RDD)

- Exploits a cutoff threshold that determines eligibility for treatment.
- Units just above and below the cutoff are compared, as they are assumed to be similar in all other respects.

Example: Students scoring just above a test score cutoff receive a scholarship; those just below do not. Comparing outcomes for students near the threshold isolates the causal effect of the scholarship award.

A photograph showing a person's hands pointing at several sheets of paper on a wooden table. The papers contain various data visualizations, including bar charts, line graphs, and donut charts. The person is wearing a brown sleeve and a white watch. The background is a plain, light-colored wall.

Additional Quasi-Experimental Designs

Further Approaches

- Beyond DiD and RDD, researchers use additional quasi-experimental methods to approximate causal effects when randomization is not feasible.

Three Key Designs

- 1. Interrupted Time Series:** tracks outcome trends before and after an intervention (e.g., childcare enrollment after a subsidy program launch).
- 2. Matching Designs:** pairs treated individuals with similar untreated counterparts (e.g., families receiving housing assistance matched with comparable controls).
- 3. Natural Experiments:** exploit external events that create treatment differences (e.g., policy changes affecting some communities but not others).

Case Study: Moving to Opportunity Experiment

Research Question

- Does moving low-income families to lower-poverty neighborhoods improve their long-term outcomes?
- Ludwig et al. (2013) examined this question using a landmark randomized experiment.

Study Design: Three Groups

- Experimental group: housing voucher + counseling + required move to low-poverty area
- Section 8 voucher group: standard housing. voucher with no location requirement.
- Control group: no voucher, remained in original public housing



Why MTO Is an Experimental Design

1

- Randomization creates a credible counterfactual.
- Treatment and control groups are statistically equivalent before the intervention begins.

2

- Groups were similar before the intervention, ruling out pre-existing differences as an explanation for outcome differences.

3

- Findings: improved adult mental health, meaningful neighborhood changes, but limited long-term economic effects.

MTO demonstrates how experimental design can isolate the causal impact of social environments on human outcomes.

Choosing Your Research Design

Step 1

- **Can the treatment be randomized?**
- Ask whether ethical, practical, and logistical conditions allow random assignment.



Yes → Experimental Design.
Randomly assign units to treatment and control groups to ensure valid causal inference.

No → Quasi-Experimental.
Use natural variation in policies, timing, or geography to construct a credible comparison group.

Step 2

- **Is there a credible comparison group?**
- Good causal designs require:
 - ✓ Clear treatment definition
 - ✓ Appropriate comparison group
 - ✓ Correct timing
 - ✓ Assumptions that support causal interpretation

Next Class

- Experimental and Quasi-experimental methods continuation
- Problem set due Sunday, 8/16 by 11:59pm

Questions?

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15-minute break